

Signature (4,4) Einstein-Spinor Field Equations in Component Form on Coordinates $\{\mathbf{x}_0, \dots, \mathbf{x}_7\}$

Complete coordinate-explicit provenance report

Reproducible exact-real implementation

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Abstract

This report audits and rewrites the curved $\text{Spin}(4,4)$ Einstein-spinor model in the coordinate ordering

`coordinates = {x0, x1, x2, x3, x4, x5, x6, x7}`

The expression occurs verbatim in the external `gpt-5.6_bridge.nb` reference notebook. Its full path and audited SHA-256 are frozen in the machine-readable evidence file. Only the coordinate names and ordering are adopted. The reference notebook's different frame, metric, and bridge connection are not imported into this model.

The matter field is one commuting real 16-component spinor field $\text{psi}=(\text{psi1},\dots,\text{psi16})^T$. There is no separate quintessence scalar and no cosmological constant. All dark-sector-like behavior comes from the nonlinear spinor potential. This report first states the covariant field theory and then lists every independent Einstein equation and all 16 spinor equations after the homogeneous ansatz used by the numerical study. It does not claim that the homogeneous list is the unrestricted eight-coordinate PDE system.

1 Scope, coordinates, and signature

The authoritative machine-readable audit is `refinement/phase7-x0-x7/evidence.json`. It independently checks seven exact component identities, four action-and-Einstein identities, seven solver-source claims, and 22 numerical-output claims.

Coordinate and tangent indices are

$$\mu, \nu, \rho, \sigma, a, b \in \{0, 1, 2, 3, 4, 5, 6, 7\}.$$

The proper evolution coordinate is `x4`; a dot means $d/dx4$. The transverse index set is

$$I, J \in \{0, 1, 2, 3, 5, 6, 7\}.$$

Tangent-signature matrix:

$$\eta_{ab} = \text{diag}(1, 1, 1, 1, -1, -1, -1, -1).$$

The homogeneous coframe, inverse frame, and metric are

$$e_\mu^a = \text{diag}(a, a, a, a, 1, a, a, a),$$

$$e_a^\mu = \text{diag}(a^{-1}, a^{-1}, a^{-1}, a^{-1}, 1, a^{-1}, a^{-1}, a^{-1}),$$

$$g_{\mu\nu} = \text{diag}(a(x4)^2, a(x4)^2, a(x4)^2, a(x4)^2, -1, -a(x4)^2, -a(x4)^2, -a(x4)^2).$$

$$H = \frac{\dot{a}}{a}.$$

Thus a constant- $\mathbf{x4}$ slice has signature $(4,3)$, not a positive-definite seven-space. This is a split-signature mathematical cosmology, not an ordinary Lorentzian one-time cosmology.

2 Covariant field theory before symmetry reduction

The real commuting spinor is a section of the rank-16 real bundle associated to $\mathbf{Spin}(4,4)$. With

$$D_\mu \psi = \partial_\mu \psi + \frac{1}{8} \omega_{\mu ab} [\gamma^a, \gamma^b] \psi, \quad \bar{\psi} = \psi^\top C,$$

the action is

$$I[g, \psi] = \int d^8 x \sqrt{|g|} \left[\frac{R}{2\kappa_8} + \frac{1}{2} \left(\bar{\psi} \gamma^\mu D_\mu \psi - (D_\mu \bar{\psi}) \gamma^\mu \psi \right) - V(S) \right],$$

where $\mathbf{S}=\mathbf{bar}(\mathbf{psi}) \mathbf{psi}$. There is no scalar field and no cosmological-constant term. Variation gives the covariant equations

$$G_{\mu\nu} = \kappa_8 T_{\mu\nu}, \quad (\gamma^\mu D_\mu - V_S) \psi = 0,$$

with

$$T_{\mu\nu} = -\frac{1}{4} [\bar{\psi} \gamma_\mu D_\nu \psi + \bar{\psi} \gamma_\nu D_\mu \psi - (D_\mu \bar{\psi}) \gamma_\nu \psi - (D_\nu \bar{\psi}) \gamma_\mu \psi] + g_{\mu\nu} \mathcal{L}_\psi.$$

These are the unrestricted PDEs. Sections 4 through 7 impose $\mathbf{psi}=\mathbf{psi}(\mathbf{x4})$ and the metric of Section 1; only that reduction is integrated.

3 Exact spinor components, bilinear, and potential

Write the mathematical components one-based as

$$\psi = (\psi_1, \dots, \psi_{16})^\top.$$

The implementation and CSV are zero-based:

$\psi_A \text{ in this report} = \mathbf{psi}(A-1) \text{ in code and CSV.}$

The canonical time generator from `cl44-seed.json` is exactly

$$\gamma^4 = \begin{pmatrix} 0 & I_8 \\ -I_8 & 0 \end{pmatrix}, \quad (\gamma^4)^2 = -I_{16}.$$

The symmetric bilinear matrix obeys

$$C^\mathsf{T} = C, \quad C^2 = I_{16}, \quad (\gamma^4)^\mathsf{T} C + C \gamma^4 = 0,$$

and has eigenvalue signature $(8, 8)$. Its condensate is fully expanded as

$$S = 2(\psi_1\psi_{16} + \psi_2\psi_{15} - \psi_3\psi_{14} - \psi_4\psi_{13} \\ + \psi_5\psi_{12} + \psi_6\psi_{11} - \psi_7\psi_{10} - \psi_8\psi_9).$$

The implemented real branch is the positive-condensate cone $S > 0$. The potential and its derivative are

$$V(S) = \frac{1}{20}S + \frac{19}{20}S^{1/5}, \quad U(S) := V_S = \frac{1}{20} + \frac{19}{100}S^{-4/5}.$$

The derivative $U(S)$ diverges as S approaches zero from above. The solver therefore rejects $S \leq 0$; the canonical finite interval remains strictly in $S > 0$.

4 Homogeneous connection, Einstein tensor, and stress tensor

The 21 nonzero Levi-Civita coefficients are

$$\Gamma^4_{00} = \Gamma^4_{11} = \Gamma^4_{22} = \Gamma^4_{33} = a\dot{a}, \\ \Gamma^4_{55} = \Gamma^4_{66} = \Gamma^4_{77} = -a\dot{a}, \\ \Gamma^I_{4I} = \Gamma^I_{I4} = H \quad (I = 0, 1, 2, 3, 5, 6, 7).$$

Define

$$Q = 6\dot{H} + 21H^2.$$

The nonzero Einstein-tensor components are

$$G_{00} = G_{11} = G_{22} = G_{33} = -a^2Q, \quad G_{44} = 21H^2,$$

$$G_{55} = G_{66} = G_{77} = a^2Q.$$

All off-diagonal $G_{\mu\nu}$ vanish under this ansatz. On shell,

$$\rho = V(S) = \frac{1}{20}S + \frac{19}{20}S^{1/5},$$

$$p = SV_S - V = -\frac{19}{25}S^{1/5}.$$

Consequently,

$$T_{00} = T_{11} = T_{22} = T_{33} = a^2 p, \quad T_{44} = \rho,$$

$$T_{55} = T_{66} = T_{77} = -a^2 p,$$

and all off-diagonal stress components vanish. These formulas, including every sign, are independently reconstructed with exact rational arithmetic by `scripts/check_einstein_spinor_model.py`.

5 All 36 independent homogeneous Einstein equations

Use

$$G_{\mu\nu} = \kappa_8 T_{\mu\nu}, \quad \kappa_8 = 21.$$

The eight diagonal equations are

$$\begin{aligned} E_{00} : & \quad -a^2 Q = 21a^2 p, \\ E_{11} : & \quad -a^2 Q = 21a^2 p, \\ E_{22} : & \quad -a^2 Q = 21a^2 p, \\ E_{33} : & \quad -a^2 Q = 21a^2 p, \\ E_{44} : & \quad 21H^2 = 21\rho, \\ E_{55} : & \quad a^2 Q = -21a^2 p, \\ E_{66} : & \quad a^2 Q = -21a^2 p, \\ E_{77} : & \quad a^2 Q = -21a^2 p. \end{aligned}$$

For $a > 0$, these reduce to the two scalar relations

$$\boxed{H^2 = \rho}, \quad \boxed{6\dot{H} + 21H^2 = -21p}.$$

Define $E_{\mu \nu} = G_{\mu \nu} - 21 T_{\mu \nu}$. The 28 independent off-diagonal equations are each zero:

$$\begin{aligned} E_{01} &= E_{02} = E_{03} = E_{04} = 0, \\ E_{05} &= E_{06} = E_{07} = 0, \\ E_{12} &= E_{13} = E_{14} = E_{15} = 0, \\ E_{16} &= E_{17} = 0, \\ E_{23} &= E_{24} = E_{25} = E_{26} = E_{27} = 0, \\ E_{34} &= E_{35} = E_{36} = E_{37} = 0, \\ E_{45} &= E_{46} = E_{47} = 0, \\ E_{56} &= E_{57} = E_{67} = 0. \end{aligned}$$

The reflected equations $E_{\nu \mu}=0$ follow from symmetry. This is the full 36-component symmetric Einstein system after the homogeneous ansatz, not the general component form for an arbitrary eight-dimensional metric and spinor.

6 Spinor field equations in complete component form

Homogeneous Dirac equation:

$$\dot{\psi} = -\frac{7}{2}H\psi - U\gamma^4\psi.$$

Using the exact γ^4 above, the first eight scalar equations are

$$\begin{aligned}\dot{\psi}_1 &= -\frac{7}{2}H\psi_1 - U\psi_9, & \dot{\psi}_2 &= -\frac{7}{2}H\psi_2 - U\psi_{10}, \\ \dot{\psi}_3 &= -\frac{7}{2}H\psi_3 - U\psi_{11}, & \dot{\psi}_4 &= -\frac{7}{2}H\psi_4 - U\psi_{12}, \\ \dot{\psi}_5 &= -\frac{7}{2}H\psi_5 - U\psi_{13}, & \dot{\psi}_6 &= -\frac{7}{2}H\psi_6 - U\psi_{14}, \\ \dot{\psi}_7 &= -\frac{7}{2}H\psi_7 - U\psi_{15}, & \dot{\psi}_8 &= -\frac{7}{2}H\psi_8 - U\psi_{16}.\end{aligned}$$

The remaining eight are

$$\begin{aligned}\dot{\psi}_9 &= -\frac{7}{2}H\psi_9 + U\psi_1, & \dot{\psi}_{10} &= -\frac{7}{2}H\psi_{10} + U\psi_2, \\ \dot{\psi}_{11} &= -\frac{7}{2}H\psi_{11} + U\psi_3, & \dot{\psi}_{12} &= -\frac{7}{2}H\psi_{12} + U\psi_4, \\ \dot{\psi}_{13} &= -\frac{7}{2}H\psi_{13} + U\psi_5, & \dot{\psi}_{14} &= -\frac{7}{2}H\psi_{14} + U\psi_6, \\ \dot{\psi}_{15} &= -\frac{7}{2}H\psi_{15} + U\psi_7, & \dot{\psi}_{16} &= -\frac{7}{2}H\psi_{16} + U\psi_8.\end{aligned}$$

There are no hidden matrix sums in these equations. Differentiating the exact condensate and using $(\gamma^4)^T C + C \gamma^4 = 0$ cancels every U term:

$$\dot{S} = -7HS.$$

Together with $S(0)=1$ and $a(0)=1$, this gives the exact background identity

$$\boxed{S = a^{-7}}.$$

7 Reduced autonomous ODE system solved by the study

The state vector and the remaining two equations are

$$y = (a, H, \psi_1, \dots, \psi_{16}) \in \mathbb{R}^{18}.$$

$$\dot{a} = aH, \quad \dot{H} = -\frac{21}{6}(\rho + p).$$

Together with the 16 explicit equations in Section 6, these form the complete implemented 18-state right-hand side. The Friedmann equation $H^2 = \rho$ is an initial constraint and an independently monitored numerical invariant, not a nineteenth evolution equation.

The reference data are imposed at $x_4=0$, not at the first CSV row:

$$a(0) = 1, \quad H(0) = 1, \quad \psi_1(0) = 1, \quad \psi_{16}(0) = \frac{1}{2},$$

with every other spinor component zero. Therefore

$$S(0) = 1, \quad \rho(0) = 1, \quad p(0) = -\frac{19}{25}, \quad H(0)^2 = \rho(0).$$

CVODE integrates one branch backward from $x_4=0$ to -0.2 and another forward from 0 to 1.5 . The backward samples are reversed, the duplicate reference sample is removed, and the resulting CSV has 171 ordered samples at spacing 0.01 .

8 Exhaustive dark-sector classification within this action and ansatz

The word “all” is bounded here. The implemented potential contains exactly two monomials, so there are exactly two homogeneous effective-fluid contributions within this action and ansatz:

1. Linear term:

$$\rho_d = \frac{1}{20}S = \frac{1}{20}a^{-7}, \quad p_d = 0, \quad w_d = 0.$$

It is dust-like at homogeneous background level and is therefore a candidate dark-matter analogy only.

1. Fractional self-interaction:

$$\rho_n = \frac{19}{20}S^{1/5} = \frac{19}{20}a^{-7/5}, \quad p_n = -\frac{4}{5}\rho_n, \quad w_n = -\frac{4}{5}.$$

It is a negative-pressure, dark-energy-like homogeneous component. For seven transverse dimensions,

$$\frac{\ddot{a}}{a} = H^2 + \dot{H} = -\frac{5}{2}\rho - \frac{7}{2}p,$$

so acceleration requires

$$w < -\frac{5}{7}.$$

The fractional term satisfies this inequality. For the mixture, the analytic transition scale is

$$a_* = \left[\frac{5(1/20)}{(2 - 7/5)(19/20)} \right]^{1/(7(4/5))} = 0.8631436165767085 \dots,$$

and the canonical interpolation gives $x_4 = -0.13800529330324499 \dots$.

These statements do not establish clustering, perturbative stability, compactification to a one-time Lorentzian theory, particle identity, or an observational fit. Infinitely many other potentials and nonminimal couplings are possible and are outside this exhaustive classification of the implemented two-term model.

9 Verification evidence and reproduction

The exact component verifier checks the signed-permutation structures of `gamma4` and `C`, $(\text{gamma4})^2 = -I$, $C^T = C$, $C^2 = I$, and skew-adjointness. The evidence builder additionally runs the exact action/Einstein checker and all 22 canonical numerical checks. Its canonical SHA-256 is 58373d6aeea90807a3f064f395b8367502ff3a280050a5c8823c9e0cc0b2eb0b.

9.1 Windows PowerShell

```
python refinement\phase7-x0-x7\verify_component_claims.py
python refinement\phase7-x0-x7\build_evidence.py
.\scripts\verify_phase7_x0_x7_reports.ps1
```

9.2 Git Bash or WSL

```
python.exe refinement/phase7-x0-x7/verify_component_claims.py
python.exe refinement/phase7-x0-x7/build_evidence.py
bash ./scripts/verify_phase7_x0_x7_reports.sh
```